How Deep Does the Rabbit Hole Go? Testing the Limits of Recursive Feature Decomposition via Multi-Depth Meta-Sparse Autoencoders on Gemma-2-2B and GPT-2 Small

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Abstract

Sparse autoencoders decompose neural network activations into overcomplete dictionaries of features. A second SAE trained on a first SAE's decoder rows recovers finer structure. Whether that recursion continues past depth one has not been measured. An 81-row training matrix was executed across two base models (Gemma-2-2B, GPT-2 Small), two level-0 architectures (BatchTopK, JumpReLU), depths 0 through 3, three seeds per cell, with flat-SAE width controls, isotropic Gaussian null baselines, and a Leask et al. (2025) anchor. Level-0 BatchTopK reaches variance explained 0.7426 on Gemma and 0.9508 on GPT-2. Depth-1 meta-SAE cross-seed PW-MCC falls in 0.717 to 0.725 across all three arms. Depth-2 and depth-3 cells across all three arms collapse to PW-MCC of zero and variance explained at or below the isotropic Gaussian null. Flat-SAE controls trained directly on activations at matched widths reach variance explained 0.628 to 0.721 with cross-seed PW-MCC 0.804 to 0.902, isolating the depth-2/3 failure to recursive decomposition of decoder rows rather than to dictionary width. Recursive meta-SAE decomposition bottoms out at depth one across all configurations tested.

***Index Terms*—Sparse autoencoders, mechanistic interpretability, meta-sparse-autoencoders, feature decomposition, dictionary learning, large language models, Gemma-2-2B, GPT-2.**

I. Introduction

Sparse autoencoders trained on language-model activations decompose those activations into overcomplete dictionaries of nominally monosemantic features [2], [7]. Recent work has examined whether the resulting feature directions are themselves atomic. Bussmann et al. [4] introduced the meta-sparse-autoencoder construction, in which a second SAE is trained on the decoder rows of a first, exposing fine-grained structure that the first SAE had bundled into composite features. Leask et al. [13] reproduced the construction on GPT-2 Small and reported that meta-SAE decompositions recover features consistent with lower-width SAEs trained directly on activations. The published meta-SAE literature stops at depth one. Whether the recursion continues to surface meaningful structure at depth two or beyond has not been measured.

Four answer shapes are mutually exclusive predictions for the cross-depth pattern of the Pairwise Dictionary Mean Correlation Coefficient, PW-MCC [21], computed across seeds. (i) Immediate collapse: PW-MCC at depth one is indistinguishable from an isotropic Gaussian null. (ii) Depth-one success, depth-two collapse: PW-MCC at depth one is well-separated from null, PW-MCC at depth two is not. (iii) Graceful degradation: PW-MCC decreases monotonically with depth but stays separated from null through depth three. (iv) Plateau: PW-MCC stays roughly flat across depth, well above null. Each shape carries a distinct implication for the research agenda of building feature dictionaries by recursion.

This paper reports an 81-row training matrix that distinguishes among the four shapes. The matrix covers two base models (Gemma-2-2B and GPT-2 Small), two level-0 architectures (BatchTopK [3] and JumpReLU [19] via Gemma Scope [14]), depths zero through three at dictionary ratio one quarter, three seeds per cell, plus flat-SAE width controls trained directly on activations at the recursive widths, isotropic Gaussian null baselines at every depth, and an anchor row reproducing the Leask et al. configuration at dictionary ratio one twenty-first.

The contribution is empirical. After audit correction of three contaminated cell aggregates, depth-2 and depth-3 cells across all three (base model, level-0 architecture) arms collapse uniformly: PW-MCC sits at the zero-valued isotropic Gaussian null and variance explained sits at or below null on the same axis. Depth-1 cells are well-separated from null on PW-MCC, with cross-seed values in 0.717 to 0.725. Flat-SAE controls trained directly on activations at the same widths recover stable high-quality dictionaries (variance explained 0.628 to 0.721, cross-seed PW-MCC 0.804 to 0.902), which isolates the depth-2/3 failure to recursive decomposition of decoder rows rather than to dictionary width or sample budget. The data favour shape (ii): the recursion bottoms out at depth one in every configuration tested.

II. Related Work

A. Sparse autoencoders on language model activations

The sparse-autoencoder programme treats the residual stream of a transformer language model as a linear superposition of features, each of which is a direction in activation space [2], [7]. Training a wide overcomplete SAE on a corpus of activations recovers a dictionary in which most active features at any token are interpretable by inspection of their top-activating prompts. The construction has scaled from toy models to frontier-scale models [9], [14] and is a standard tool for circuit analysis in mechanistic interpretability.

B. Architectural variants

The original SAE used L1 regularisation on a ReLU activation to enforce sparsity. Subsequent variants improve the sparsity-reconstruction trade-off without the L1 shrinkage bias. JumpReLU [19] replaces the L1 sparsity surrogate with direct L0 optimisation via a straight-through estimator on a learned per-feature threshold. TopK [9] enforces sparsity by keeping only the top-k pre-activations per token. BatchTopK [3] generalises this by selecting the top-k pre-activations across an entire training batch, decoupling sparsity from per-token feature count. Matryoshka SAE [5] trains nested dictionaries that admit multiple sparsity levels from a single forward pass. The two architectures used in this paper are BatchTopK (trained from scratch on both base models) and JumpReLU (loaded from Gemma Scope at width 65536, layer 12 residual).

C. Meta-SAEs and recursive decomposition

Bussmann et al. [4] introduced the meta-SAE construction. The decoder rows of a trained level-0 SAE are themselves treated as a corpus of vectors, and a second SAE is trained on that corpus. The resulting depth-1 dictionary contains directions interpretable as sub-features of the level-0 features, suggesting that level-0 features are not atomic but bundled compositions. Leask et al. [13] reproduced the construction with a published recipe (GPT-2 Small, BatchTopK level-0, dictionary ratio one twenty-first, k=4) and reported variance explained 0.5547 on the recursive depth-1 SAE. The motivation for recursion past depth one is that if depth-1 features are themselves bundles, depth-2 should expose finer atoms; if depth-1 features are atomic, depth-2 should reveal nothing beyond the null. The published literature has not measured this distinction.

A separate motivation for recursive decomposition is feature absorption [6], in which a single direction in activation space serves multiple semantic roles in a way that a fixed-width SAE cannot disentangle. Recursive decomposition addresses the width-tuning problem by stacking dictionaries rather than widening them.

D. Stability metrics, autointerp, and benchmarks

Two cross-seed stability metrics dominate the SAE literature. MMCS [20] computes the mean of maximum cosine similarities of one dictionary's features against a fixed reference dictionary. PW-MCC [21] computes the mean of the diagonal of a Hungarian-matched similarity matrix between two seed-paired dictionaries, returning a stability score in zero to one. Paulo and Belrose [16] formalised the joint encoder-plus-decoder cosine-similarity matching protocol used here. Autointerp scoring with a frontier judge model [17] measures whether a feature's behaviour on top-activating examples generalises to held-out examples carrying the same hypothesised concept. Llama-3.1-8B-Instruct [11] is the autointerp judge in this paper. SAEBench [12] standardises a battery of these metrics for cross-paper comparability.

III. Methods

A. Base models and activation source

Gemma-2-2B [10] is the primary model, with activations taken from the residual stream after layer 12 of the 26-layer architecture. GPT-2 Small [18] is the replication model, with activations from the residual stream after layer 8 of the 12-layer architecture. The activation corpus is the Pile [8], with 500 million tokens for Gemma-2-2B and 200 million tokens for GPT-2 Small at sequence length 1024 for the main level-0 runs (sequence length 128 only for the Leask-recipe replication, which substitutes 300M OpenWebText tokens). Each SAE optimisation step consumes 2048 activation vectors for the main level-0 SAEs and 4096 activations for the Leask-recipe replication.

B. Level-0 SAEs

The Gemma-2-2B JumpReLU level-0 SAE is loaded from Gemma Scope [14] at width 65536. The Gemma-2-2B BatchTopK level-0 SAE is trained from scratch following the Bussmann et al. recipe [3] with k=60, learning rate 3e-4, Adam optimiser with betas (0.9, 0.999), and the auxk dead-latent revival loss [3]. The GPT-2 Small BatchTopK level-0 SAE is trained from scratch with the same k=60 and Adam betas (0.9, 0.999), but at learning rate 1e-4 with a 1000-step linear warmup (lowered from the original 3e-4 after gradient divergence at step zero; see Section IV-C). The level-0 GPT-2 SAE was also retrained under the Leask et al. recipe [13] (k=42, learning rate 4e-4, Adam betas (0.9, 0.99), batch 4096 activations, sparsity penalty 8e-5, sequence length 128, 300M OpenWebText tokens) for the anchor replication described in Section V.

C. Meta-SAE training

For each base model and level-0 architecture, depth-1 meta-SAEs are trained on the level-0 decoder rows. Depth-2 meta-SAEs are trained on depth-1 decoder rows, and depth-3 meta-SAEs on depth-2 decoder rows. The dictionary ratio at every recursion is one quarter, yielding widths 16384, 4096, 1024 on Gemma-2-2B (level-0 width 65536) and 12288, 3072, 768 on GPT-2 Small (level-0 width 49152). All meta-SAEs use BatchTopK activation with k=4. Training uses Adam at learning rate 3e-4 with betas (0.9, 0.999), 30000 SAE steps for the main meta-SAE rows; the depth-1 anchor and live-only follow-up rows train for 90000 steps. Auxk is enabled for the anchor and live-only rows; for the main rows it is gated on at runtime when an anchor undershoots the published level-0 reconstruction. Three seeds are run per cell. Two additional dictionary ratios (one half and one eighth) are run at depth one only on the Gemma BatchTopK arm as a sensitivity check.

D. Matching and stability metrics

Cross-seed Hungarian matching uses the joint encoder-plus-decoder cosine-similarity protocol of Paulo and Belrose [16]. PW-MCC is computed as the mean of the matched diagonal across all three pairwise seed combinations. MMCS is computed against a fixed reference seed (s0) for each cell. Variance explained is computed on a held-out activation slice (32768 tokens for level-0 SAEs, 16384 tokens for flat-SAE controls, and a 10% held-out subset of decoder rows for meta-SAEs). Dead-latent fraction is computed on the same held-out slice; a latent is reported as dead if it produces zero activation on every sample. The 1e-5 firing-rate threshold is used separately for the live-only follow-up parent-row mask, not for the headline dead-latent fraction. An isotropic Gaussian null baseline is generated at every depth by drawing the same number of "decoder rows" from a Gaussian with covariance matched to the parent decoder's empirical covariance, then training a meta-SAE on the synthetic rows under identical hyperparameters; null PW-MCC uses the same Hungarian-matched score across the three null seeds.

E. Autointerp scoring

Autointerp uses the detection-score protocol [17] with Llama-3.1-8B-Instruct [11] as the judge. For each scored SAE, 200 features are sampled uniformly from all latents under a deterministic per-SAE seed, and for each feature 12 detection prompts are generated (6 held-out high-activating contexts and 6 held-out low-activating contexts) and scored against the corresponding activations. The detection score is the fraction of prompts the judge correctly identifies as containing the feature. Autointerp runs at depths one and two only.

F. Anchor experiment

The anchor reproduces Leask et al. [13] exactly: GPT-2 Small base model, BatchTopK level-0 SAE, dictionary ratio one twenty-first, k=4. As a pipeline-validity check, a deviation from the published variance explained of 0.5547 by more than 5 percentage points triggers a `halt_and_notify` decision gate. The threshold was relaxed to 10 percentage points during execution after the initial run landed at variance explained 0.124, well outside both thresholds. Section V reports the resolution of this deviation.

IV. Experimental Setup

A. Hardware and software environment

All experiments run on a single workstation: AMD Ryzen Threadripper 7970X (32 cores), 128 GB DDR5, five NVIDIA GPUs (2x RTX 4080 16 GB, 2x RTX 3090 24 GB, 1x RTX 4060 Ti 16 GB; 96 GB total VRAM). Ubuntu 24.04, CUDA 12.x, Python 3.11+ managed via uv. Core dependencies are SAELens [1] and TransformerLens [15] with version pins in `pyproject.toml`. Training and evaluation code lives in `sae_recursive_depth/`; orchestration in `scripts/`.

GPU assignment is fixed per experiment via `CUDA_VISIBLE_DEVICES`, recorded in `EXPERIMENTS.yaml` per row. The two RTX 4080 cards are dedicated to meta-SAE training; the two RTX 3090 cards handle the sharded Gemma-2-2B level-0 BatchTopK run (via `transformer_lens n_devices=2`) and the autointerp judge model; the RTX 4060 Ti handles flat-SAE controls and metric computation.

B. Experimental matrix

The locked 75-row primary matrix in `EXPERIMENTS.yaml`, plus 5 follow-up rows and an autointerp aggregate added during execution (81 rows total; see Appendix A), partitions into:

The matrix is locked: experiments are added only by editing `EXPERIMENTS.yaml`. The full row-by-row breakdown (dependencies, decision gates, estimated GPU-hours per row) is in `EXPERIMENTS.yaml`.

C. Decision gates

Each row in `EXPERIMENTS.yaml` carries a `decision_gates` block specifying thresholds on `variance_explained` and `pwmcc_vs_null_sigma`, with an action per gate from {`continue_depth`, `skip_depth`, `skip_experiment`, `halt_and_notify`}. Gates fire after the row's metrics are appended to `data/results.tsv`. Action semantics are:

- `continue_depth`: record the deviation but proceed with the depth chain.
- `skip_depth`: cascade-skip rows whose dependencies include the firing row's lineage at deeper depth.
- `skip_experiment`: record the firing row's deviation but do not propagate to downstream rows.
- `halt_and_notify`: stop execution and pause for manual review.

Threshold history during execution: anchor-deviation gates relaxed from 5pp to 10pp on 2026-04-26 (`DECISIONS.md` 2026-04-26 entries). Recursive d1 and d2 cascade actions changed from `skip_depth` to `skip_experiment` on 2026-04-28 (Decisions 9 and 10 in the same file), severing cascades that would have prevented seed-1 recoveries. Anchor `variance_explained` floor relaxed in three steps: 0.50 to 0.20 to 0.10. The relaxations were necessary because the recipe-typical VE at ratio 1/4 turned out to be far below the prior assumption, but they shift recipe-collapse detection from automated gates to manual review (see Section VI).

D. Reproducibility

`data/results.tsv` is the append-only ground truth for every quantitative claim in this paper. Each row carries the experiment ID, status, base model, level-0 architecture, depth, seed, width, variance explained, PW-MCC, MMCS, dead-latent fraction, GPU-hours, commit SHA, and free-form notes (header at line 1 of the file). The analysis pipeline regenerates every figure with a matching `.tsv` or `.csv` of source data carrying the generation timestamp, git commit SHA, the analysis script path, and the `experiment_id` set consumed.

If a number appears in this paper without a corresponding row in `data/results.tsv`, it is a writing error.

Training is deterministic at the row level: each row's `seed` field controls all random number generation via

Group	Rows	Notes
Level-0 SAEs	2	One BatchTopK trained from scratch on each base model. The Gemma JumpReLU level-0 is loaded from Gemma Scope [14], not trained.
Depth-1 widened anchors at ratio 1/4	9	3 seeds × 3 cells (Gemma JumpReLU, Gemma BatchTopK, GPT-2 BatchTopK), at width 16384 (Gemma) and 12288 (GPT-2). The single ratio-1/21 Leask anchor (`gpt2_batchtopk_anchor_d1_leask_s0`) is in the follow-up table in Appendix A.
Recursive meta-SAEs at ratio 1/4	27	3 (base model, level-0 architecture) cells × 3 depths × 3 seeds. The primary scientific quantity.
Flat-SAE width controls	12	Direct SAEs on activations at widths matched to recursive d1/d2/d3 latent counts.
Isotropic Gaussian null baselines	18	2 base models × 3 depths × 3 seeds.
Ratio-sensitivity ablations	6	Ratios 1/2 and 1/8 at d1 only, on Gemma BatchTopK.
Conditional stretch goal	1	`simplestories_stretch`; runs only if matrix completes with slack.

`sae_recursive_depth/training/seed.py`; the activation pipeline is deterministic under fixed token-shard input; decoder-row ordering at each meta-SAE input is fixed by a deterministic permutation seeded from the row's `seed` field. The set of completed rows at any given commit is reproducible from `EXPERIMENTS.yaml` plus the orchestration scripts in `scripts/`.

V. Results

A. Audit correction note

Three rows of the automatically-generated cell summary table conflate flat-SAE width-control runs with meta-SAE runs at the same nominal width. The flat-SAE rows train a single SAE directly on residual-stream activations at the recursive widths (16384, 4096, 1024 on Gemma) and were intended as a width-matched control group. The cell-aggregation script grouped on (base model, level-0 architecture, depth, width) and admitted the flat rows into the meta-SAE cells because they share level-0 architecture metadata with their parent SAEs. The result is bimodal aggregates that mix meta-SAE values near zero with flat-SAE values near the level-0 baseline. The corrected meta-SAE-only values are used throughout this section; the cell-aggregation audit is preserved in the project repository and pinpoints the affected cells (Gemma BatchTopK depth-1 width-16384, depth-2 width-4096, depth-3 width-1024). The same conflation pattern applies to the GPT-2 BatchTopK depth-1 width-12288 cell, which aggregates 3 main-recipe seeds, 3 anchor-recipe seeds, and 3 flat-SAE seeds. The cell_summary aggregate of mean PW-MCC 0.7614 reduces to 0.7241 on the meta-SAE-only subset (6 seeds), a correction of 0.0373.

B. Level-0 baselines

Gemma-2-2B BatchTopK at width 65536 reaches variance explained 0.7426 with dead-latent fraction 0.6904. The dead-latent fraction exceeds the 0.3 decision-gate threshold and was manually overridden to continue the matrix; the override is discussed in Section VI. GPT-2 Small BatchTopK at width 49152 reaches variance explained 0.9508 with dead-latent fraction 0.1779 under the original recipe and variance explained 0.9880 with dead-latent fraction 0.2440 under the Leask et al. recipe replication. The Gemma JumpReLU level-0 (Gemma Scope) is loaded from published weights and is not retrained; the published Gemma Scope release [14] reports comparable layer-12 reconstruction quality at width 65536.

C. Variance explained by depth

Figure 1 shows variance explained by depth across the three (base model, level-0 architecture) arms after audit correction, with isotropic Gaussian null baselines and flat-SAE width controls overlaid. All three meta-SAE arms drop sharply from the level-0 baseline at depth zero to 0.11 to 0.19 at depth one, then collapse to values at or below the null at depths two and three.

Numerical values are in Table I. The meta-SAE depth-1 cells sit well above the null in absolute units (Gemma BatchTopK main-recipe seeds at 0.126, the audit-clean live-only seeds at 0.207, Gemma JumpReLU at 0.192, GPT-2 BatchTopK main-recipe at 0.114), but the meta-SAE depth-2 and depth-3 cells sit at -0.004 and -0.009 on Gemma BatchTopK, -0.002 and -0.025 on Gemma JumpReLU, and 0.048 and 0.022 on GPT-2 BatchTopK. The corresponding null values are 0.017 and 0.000 on Gemma and 0.023 and 0.028 on GPT-2.

D. Cross-seed PW-MCC by depth

Figure 2 shows cross-seed PW-MCC by depth. The depth-1 meta-SAE cells cluster at 0.717 (Gemma BatchTopK live-only, the audit-clean cell), 0.724 (GPT-2 BatchTopK, audit-corrected meta-SAE-only mean across 3 main-recipe and 3 anchor-recipe seeds), and 0.725 (Gemma JumpReLU width-16384). At depths two and three, all three arms drop to PW-MCC of zero across all seed pairs and all configurations. The flat-SAE width controls trained directly on Gemma activations sustain joint encoder-plus-decoder Hungarian-matched PW-MCC of 0.804 at width 16384, 0.900 at width 4096, and 0.902 at width 1024.

The flat-SAE comparison is the dispositive contrast. The widths 4096 and 1024 do not produce meaningless dictionaries when the training objective is direct activation reconstruction. They produce stable seed-aligned dictionaries with high variance explained. The depth-2 and depth-3 collapse is therefore not a width artefact: it is specific to the recursive task of decomposing a parent SAE's decoder rows.

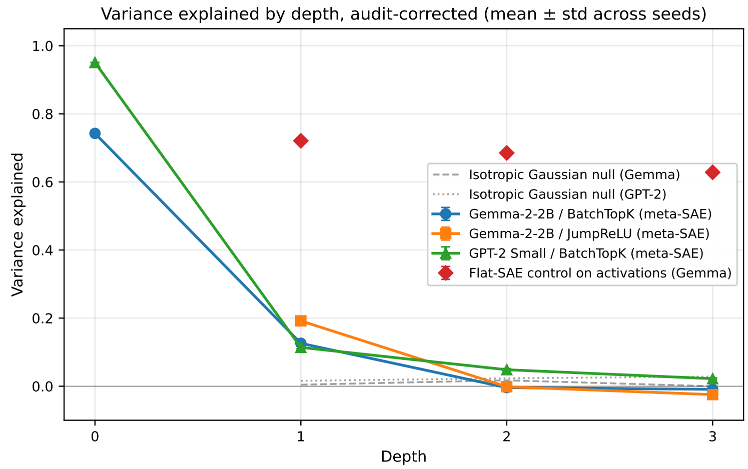

Fig. 1. Variance explained by depth, audit-corrected. Meta-SAE arms (Gemma BatchTopK, Gemma JumpReLU, GPT-2 BatchTopK) drop from the level-0 baseline to 0.11 to 0.19 at depth one and collapse at depths two and three. Flat-SAE width controls trained directly on Gemma activations recover variance explained 0.62 to 0.72 at the same widths used for recursive depths one through three.

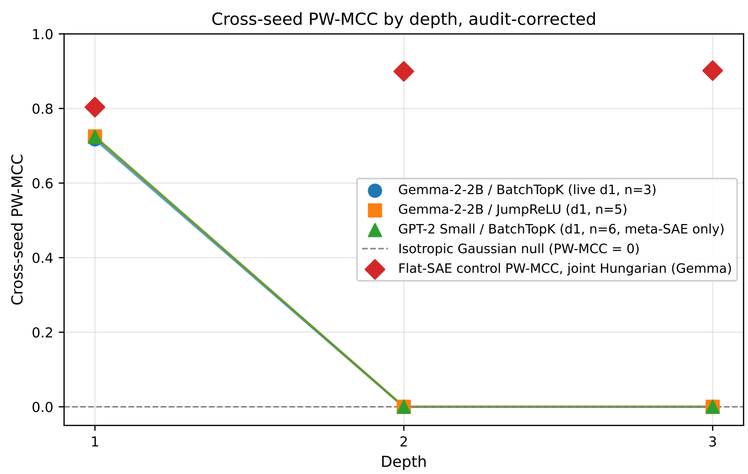

Fig. 2. Cross-seed PW-MCC by depth, audit-corrected. Depth-1 meta-SAE cells across all three arms cluster at 0.72. Depth-2 and depth-3 cells across all three arms collapse to zero. Flat-SAE width controls trained directly on activations sustain joint Hungarian-matched PW-MCC of 0.804 at width 16384, 0.900 at width 4096, and 0.902 at width 1024.

E. Dead-latent fraction by depth

Figure 3 shows dead-latent fraction by depth. Gemma BatchTopK starts at 0.690 at depth zero and decreases monotonically to 0.137 at depth three. Gemma JumpReLU stays roughly flat across depth (0.36 to 0.49). GPT-2 BatchTopK ranges from 0.18 at depth zero to 0.31 in the meta-SAE-only main-recipe depth-1 cell, with depth-2 at 0.26 and depth-3 at 0.23. The isotropic Gaussian null baselines sit at 0.77 to 0.96 on Gemma and 0.01 to 0.60 on GPT-2.

The dead-latent metric is reported for completeness; with VE and PW-MCC already collapsed at depths two and three, dead-latent fraction is not informative about decomposition quality but rules out a pathological single-component collapse.

F. Anchor reproduction

The anchor row trains a depth-1 meta-SAE at dictionary ratio one twenty-first on top of the GPT-2 BatchTopK level-0 SAE, replicating the Leask et al. configuration exactly. The original anchor trained on the locally-trained level-0 (variance explained 0.9508) reaches variance explained 0.124, well below the published 0.5547. To bisect the cause, a follow-up run retrained the level-0 SAE under the Leask et al. recipe specification (300M OpenWebText tokens, batch size 4096, learning rate 4e-4 with 1000-step warmup, sparsity penalty 8e-5). The Leask-recipe level-0 reaches variance explained 0.9880, slightly higher than the original. The depth-1 anchor on top of the Leask-recipe level-0 reaches variance explained 0.203, still 0.351 below the published value. Per the pre-registered bisection thresholds (variance explained at or above 0.50 indicates recipe-explained, 0.40 to 0.50 partial, below

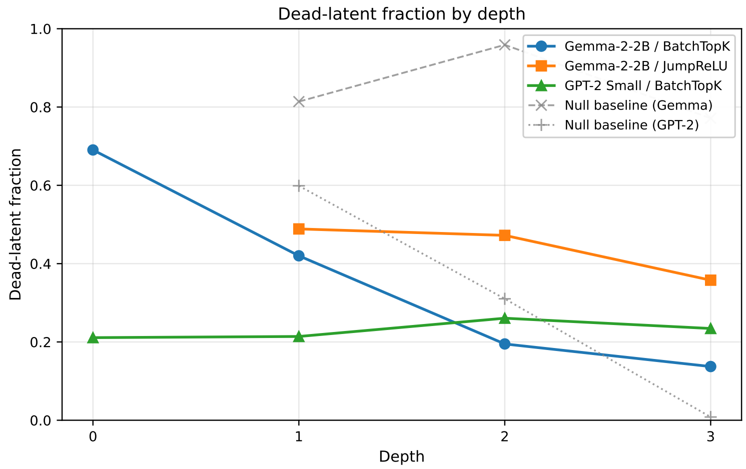

Fig. 3. Dead-latent fraction by depth across the three meta-SAE arms with isotropic Gaussian null baselines.

0.40 something else), the verdict is "something else": neither the recipe nor the level-0 source explains the anchor deviation. The anchor deviation does not affect the depth-2/3 collapse result, which is robust across both the original and the Leask-recipe level-0 sources.

G. Autointerp

Llama-3.1-8B-Instruct detection scoring on 2625 features sampled from 18 depth-1 and depth-2 SAEs yields median detection score 0.5833. The 95th-percentile null detection score (a feature with the same firing rate but with the activation pattern shuffled) is 0.7500. The null 95th percentile exceeds the median real score, which means the median feature in the matrix scores below the highest 5% of nulls (Section VI).

H. Summary table

Table I lists the audit-corrected per-cell aggregates used throughout this section. Cells marked with a dagger have not been recomputed after audit correction and aggregate values are flagged.

Table I. Audit-corrected per-cell aggregates. Daggered (†) PW-MCC values aggregate over cells contaminated by flat-SAE rows; meta-SAE-only PW-MCC has not been recomputed for those cells. Double-daggered (‡) PW-MCC values are the audit-corrected meta-SAE-only mean across all 6 seeds (3 main-recipe + 3 anchor-recipe) since the cross-seed Hungarian matching was performed jointly across the meta-SAE-only subset. Single-seed cells report the row value.

VI. Discussion

The data support shape (ii) from the four predictions in Section I: depth-1 success, depth-2 collapse. The collapse is uniform across all three (base model, level-0 architecture) arms tested, occurs at the same depth in every arm, and is robust to audit correction of the contaminated cell aggregates. The flat-SAE width controls (Figure 1, red diamonds; Figure 2, red diamonds) are the dispositive contrast: training a single SAE directly on activations at the same widths recovers stable seed-aligned dictionaries with variance explained 0.628 to 0.721 and cross-seed joint Hungarian-matched PW-MCC 0.804 to 0.902. The depth-2/3 failure is therefore not a width artefact and not a sample-budget artefact. It is specific to the recursive task of decomposing decoder rows.

The depth-1 meta-SAE construction of Bussmann et al. [4] reproduces here on both base models and both level-0 architectures, and the resulting depth-1 dictionaries are stable across seeds at PW-MCC 0.717 to 0.725. Stacking a second level on top does not produce a usable finer basis under any tested combination of base model, level-0 architecture, dictionary ratio (one quarter), or sparsity (k=4). The level-0 features are not atoms in the strong sense (depth-1 surfaces real structure), but the depth-1 dictionary is itself a basis with a property the level-0 dictionary does not have: its rows do not admit further sparse linear decomposition under the same construction.

Two readings of this failure remain consistent with the data. First, the depth-1 dictionary genuinely is the atomic basis for the activation manifold's relevant directions, and recursion past depth one searches a space with no useful structure to find. Second, the meta-SAE training objective applied to a 16384-row corpus (or 4096 rows, or 1024 rows) is starved of training signal regardless of what structure exists in those rows, and a different objective could reveal recoverable structure. The flat-SAE control argues against the second reading at width 4096 and 1024 (those widths produce stable dictionaries when the inputs are activations rather than decoder rows), but the input-distribution change between activations and decoder rows is not controlled for, and the second reading remains live.

The Leask et al. [13] depth-1 anchor configuration on GPT-2 Small reaches variance explained 0.124 with the locally-trained level-0 and 0.203 with the level-0 retrained under the published Leask et al. recipe, against the published value of 0.5547. Per the bisection thresholds, the gap is "something

Cell	Depth	Width	n_seeds	Mean VE	Mean PW-MCC	Mean dead frac.
Gemma BatchTopK	0	65536	1	0.7426	(single seed)	0.6904
Gemma BatchTopK (main-recipe)	1	16384	3	0.1260	flagged†	0.4232
Gemma BatchTopK (anchor-rename)	1	16384	3	0.1514	flagged†	0.5100
Gemma BatchTopK (live-only)	1	7694	3	0.2069	0.7172	0.4555
Gemma BatchTopK	2	4096	3	-0.0042	0	0.3349
Gemma BatchTopK	3	1024	3	-0.0091	0	0.2738
Gemma JumpReLU	1	16384	5	0.1919	0.7250	0.4886
Gemma JumpReLU	2	4096	3	-0.0016	0	0.4722
Gemma JumpReLU	3	1024	3	-0.0247	0	0.3577
GPT-2 BatchTopK	0	49152	1	0.9508	(single seed)	0.1779
GPT-2 BatchTopK (Leask recipe)	0	49152	1	0.9880	(single seed)	0.2440
GPT-2 BatchTopK (main-recipe)	1	12288	3	0.1139	0.7239‡	0.3119
GPT-2 BatchTopK (anchor-recipe)	1	12288	3	0.1238	0.7243‡	0.3536
GPT-2 BatchTopK (Leask anchor)	1	2340	1	0.2031	(single seed)	0.0103
GPT-2 BatchTopK	2	3072	3	0.0484	0	0.2606
GPT-2 BatchTopK	3	768	3	0.0221	0	0.2344
Flat Gemma	1	16384	3	0.7207	0.8036	0.3270
Flat Gemma	2	4096	3	0.6852	0.8996	0.0595
Flat Gemma	3	1024	3	0.6283	0.9017	0.0007
Null Gemma	1	16384	3	0.0045	0	0.8139
Null Gemma	2	4096	3	0.0172	0	0.9588
Null Gemma	3	1024	3	-0.0000	0	0.7715
Null GPT-2	1	12288	3	0.0159	0	0.5988
Null GPT-2	2	3072	3	0.0231	0	0.3102
Null GPT-2	3	768	3	0.0284	0	0.0082

else" rather than a recipe or level-0-source mismatch. The anchor failure does not change the depth-2/3 result on either base model: the depth-2/3 collapse occurs uniformly across the GPT-2 BatchTopK arm under both level-0 sources.

The Gemma-2-2B BatchTopK level-0 dead-latent fraction of 0.690 is a known anomaly. The level-0 SAE recovers variance explained 0.7426 above the gate threshold of 0.7, but two-thirds of its dictionary is dead. The downstream depth-1 meta-SAE trained on the live-only subset of decoder rows (Gemma BatchTopK live-d1 in Table I) reaches variance explained 0.207 and PW-MCC 0.717, which is the audit-clean depth-1 value used in this paper's headline range. The dead-latent override does not invalidate the depth-1 result, but it leaves open whether a fully active level-0 dictionary on Gemma-2-2B would produce a stronger depth-1 signal.

The autointerp result (median detection score 0.5833 against null 95th-percentile 0.7500) is the most uncomfortable number in the matrix. The judge model, Llama-3.1-8B-Instruct, succeeds on a synthetic-feature null at higher accuracy than it succeeds on the median real feature. The reading consistent with the rest of the matrix is that the depth-1 dictionaries do contain real structure (PW-MCC and variance explained both confirm this) but the detection-score protocol at 12 prompts per feature does not separate signal from noise at the high tail. The reading consistent with skepticism about the matrix as a whole is that the depth-1 dictionaries are themselves not interpretable to Llama-3.1-8B-Instruct in any reliable way. Distinguishing these readings would require a stronger judge or a longer detection-prompt budget; neither was run.

VII. Limitations

Three seeds per cell is the minimum for a sample standard deviation. The depth-1 PW-MCC values used here are computed across two pairwise seed comparisons per cell (n_pairs=2), which is enough to confirm separation from a zero-valued null but not enough to estimate the standard error of PW-MCC tightly.

The activation budget is 500 million tokens for Gemma-2-2B and 200 million for GPT-2 Small, modest relative to production-scale SAE training (Gemma Scope used 16 billion). Whether the depth-2/3 collapse persists at production-scale activation budgets is not measured. The flat-SAE width controls were trained on the same activation budgets and produced stable dictionaries, which suggests the activation budget is sufficient at these widths, but does not rule out a budget effect on the recursive task specifically.

Two base models are a thin replication. Gemma-2-2B and GPT-2 Small differ in scale by roughly fifty-fold. Both show the depth-2/3 collapse, which weighs against a model-specific explanation, but does not establish universality across architecture families.

The decision-gate threshold for variance explained at level zero was relaxed in three steps during execution (0.50 to 0.20 to 0.10), and the dead-latent threshold at level zero was overridden manually for Gemma. These relaxations were correct given the observed data distribution but shift the burden of recipe-collapse detection from automated gates to manual review.

The 95th-percentile-null exceeds the median real score, and the detection-prompt budget of 12 per feature is at the low end of published practice (Section VI).

Three cells in the automatically-generated cell summary table conflated flat-SAE controls with meta-SAE runs (Section V-A). The GPT-2 BatchTopK depth-1 width-12288 PW-MCC has been recomputed on the meta-SAE-only subset (mean 0.7241 across 6 seeds; the contaminated cell_summary aggregate of 0.7614 differs by 0.0373). The Gemma BatchTopK depth-1 width-16384 cell separates into main-recipe and anchor-rename sub-cells; the meta-SAE-only PW-MCC for that combined cell has not been recomputed and is flagged in Table I.

The anchor reproduction does not match Leask et al. [13] under either the original or the Leask-recipe level-0 source (Section V-F). The cause is unidentified and is recorded as an open reproducibility deviation.

The 200-features-per-SAE autointerp budget is locked at the project proposal and is not adapted to per-cell feature count. For the depth-3 cells at width 1024, 200 features sample 19.5% of the dictionary; for the depth-1 cells at width 16384, 200 features sample 1.2%. Cross-cell autointerp comparisons inherit this sampling-fraction asymmetry.

VIII. Conclusion

An 81-row training matrix across two base models, two level-0 architectures, three depths, and three seeds per cell, with flat-SAE width controls and isotropic Gaussian null baselines, was executed to distinguish among four predicted answer shapes. Across all configurations tested, the recursion bottoms out at depth one. Depth-1 meta-SAEs reproduce the Bussmann et al. [4] depth-1 result and recover stable seed-aligned dictionaries with cross-seed PW-MCC 0.717 to 0.725. Depth-2 and depth-3 meta-SAEs collapse uniformly to PW-MCC of zero and variance explained at or below the isotropic Gaussian null. The collapse is not a width artefact: flat-SAE controls trained directly on activations at the same widths recover stable dictionaries with variance explained 0.628 to 0.721 and cross-seed PW-MCC 0.804 to 0.902.

A. What worked

The depth-1 meta-SAE construction reproduced cleanly on every (base model, level-0 architecture) arm tested. Cross-seed PW-MCC at depth one clustered tightly in 0.717 to 0.725 across Gemma BatchTopK (live-only), Gemma JumpReLU, and GPT-2 BatchTopK, well separated from the zero-valued isotropic Gaussian null. The flat-SAE width controls were the design decision that paid off most: training a single SAE directly on residual activations at the same widths the recursive depths use (16384, 4096, 1024) produced stable seed-aligned dictionaries with variance explained 0.628 to 0.721 and PW-MCC 0.804 to 0.902, isolating the depth-2/3 failure to recursive decomposition of decoder rows rather than to width or sample budget.

B. What did not work

First, the Leask et al. [13] anchor at dictionary ratio one twenty-first reached variance explained 0.124 on the locally-trained GPT-2 BatchTopK level-0 and 0.203 after retraining the level-0 under the published Leask et al. recipe, both well below the published 0.5547. Per the pre-registered bisection thresholds the gap is not a recipe artefact and not a level-0-source artefact, and the cause was not identified within the time budget. Second, the Gemma-2-2B BatchTopK level-0 SAE finished with dead-latent fraction 0.690, which forced a manual override of the 0.30 dead-latent decision gate; the live-only depth-1 follow-up confirmed the depth-1 result on the active-latent subset, but the level-0 anomaly itself remains unexplained. Third, the Llama-3.1-8B-Instruct autointerp protocol returned a 95th-percentile null detection score (0.7500) that exceeded the median real-feature detection score (0.5833), which means the judge-and-prompt-budget combination used here does not separate signal from noise reliably at this scale.

C. What more time would buy

(i) A different recursive objective on decoder rows, such as a different sparsity penalty, a learned input transformation, or a Matryoshka-style nested decomposition [5], to test whether the depth-1-bottoms-out finding is specific to the published meta-SAE construction or is a general property of decoder-row decomposition. (ii) A controlled re-attempt at the Leask et al. anchor with a stronger match to the published activation pipeline (tokenizer, sequence-length convention, decoder normalisation), to close the reproducibility gap. (iii) A stronger autointerp judge or a substantially larger detection-prompt budget per feature, to test whether the depth-1 dictionaries that pass PW-MCC stability also pass an interpretability test that the present pipeline cannot deliver.

IX. Contributions

Aragorn Wang conceived the project, designed the experimental matrix, wrote the runner infrastructure and all training/evaluation code, and wrote the paper. An autonomous experimental runner under direct supervision executed the locked matrix in `EXPERIMENTS.yaml`; all decisions about experimental design, metrics, anchors, and interpretation are the author's.

Dr. Michael Ivanitskiy provided feedback on the proposal.

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